Name: _____

Class: _____

ACTIVITY SHEET

3.5 Relationships in ecosystems

The intrinsic relationships between organisms and their environments give rise to the level of biodiversity seen within an ecosystem. There are competitors and collaborators, symbiotic relationships and relationships with keystone species, which aid the survival of different species in different environments. Use your knowledge of the relationships present in different ecosystems to answer the following questions.

1 Define the following terms.

a Collaborator

b Competitor

c Mutualism

d Parasitism

e Commensalism

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2 Lichen is an organism consisting of two species. Generally it is a fungus and a green algae or cyanobacteria. This relationship between the fungus and algae is said to be mutualistic. Explain.

3 Explain the type of relationship that is being shown in the photograph below.



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4 Explain the type of relationship that a tick has with a kangaroo.

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5 a Explain how scientists determine whether a species is a keystone species.

b Provide an example of a keystone species and its effect on the biodiversity of an ecosystem.

6 Suggest a plausible explanation for the following data.

× = The species is present on the island.

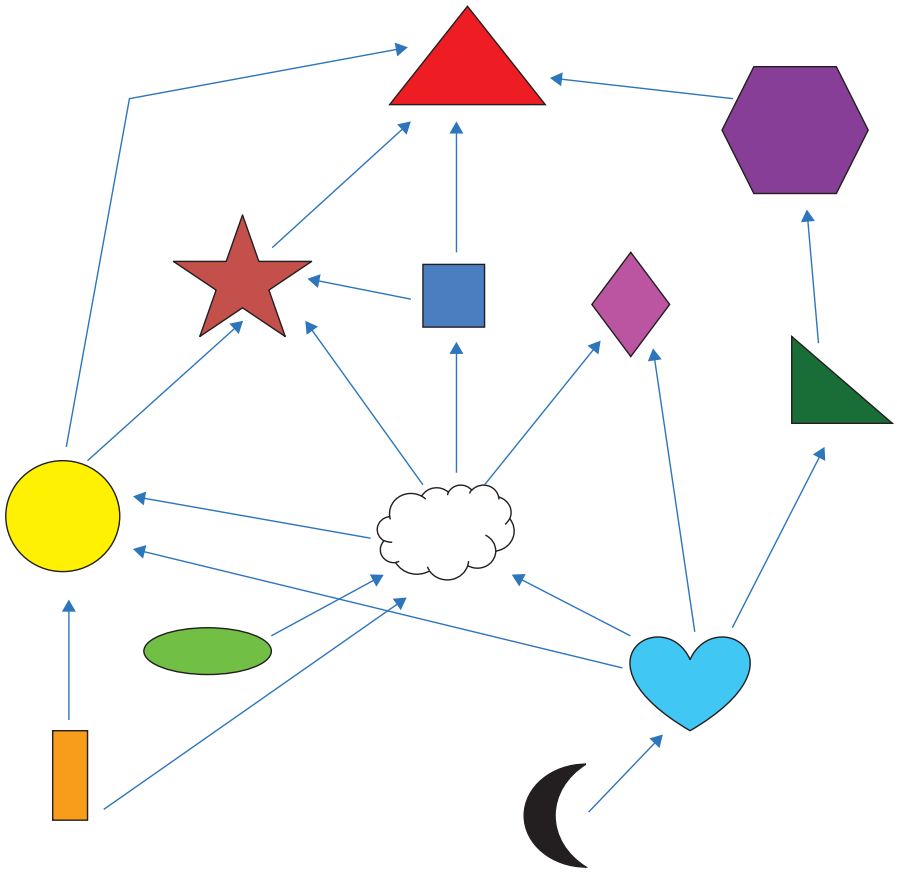
– = The species is not present on the island.

Island	Finch species 1	Finch species 2
1	×	–
2	–	×
3	–	–
4	–	–
5	–	×
6	–	×
7	×	–
8	×	–

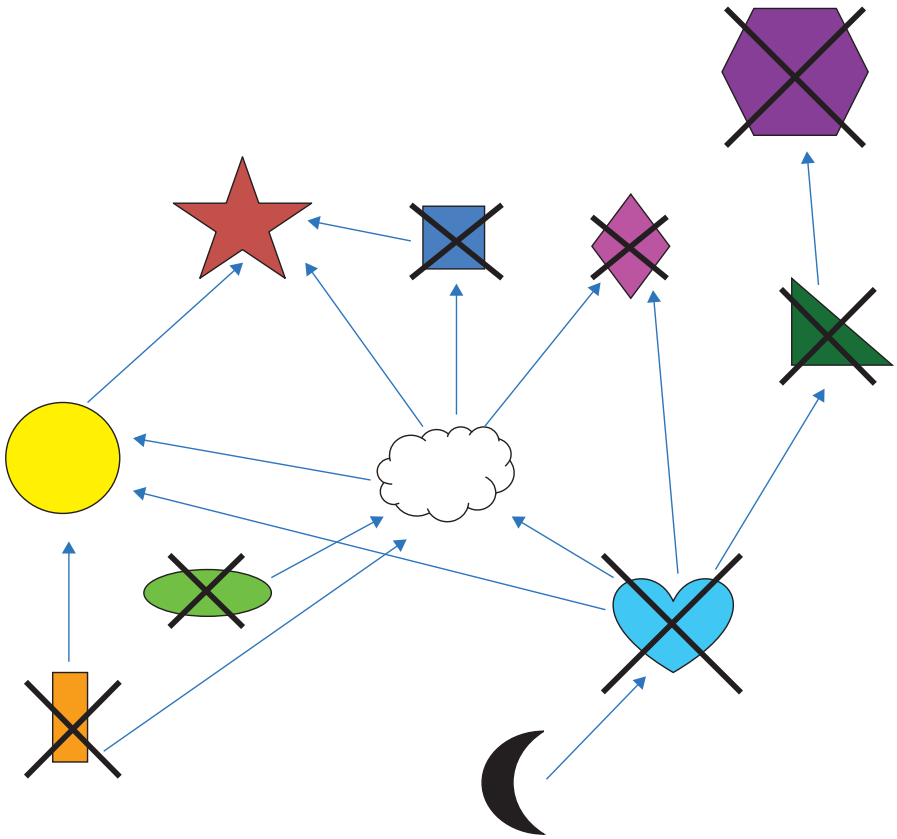
7 Cane toads were introduced to Queensland's sugarcane fields in 1935. They reproduced quickly, and the populations of native frog species started to reduce. What type of interaction is depicted here?

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8 The diagram below represents an ecological food web, where each shape represents a different species.



If you were to remove the red triangle from the food web the populations of stars and circles explode and the following occurs. Note the cross represents the species dying out.



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a Explain the role of the triangle in the food web.

b State the organism that the triangle could represent.

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